

FINAL EXPLANATION: BIOLOGY GRADE 11

Student Assessment Document

FINAL EXPLANATION: BIOLOGY GRADE 11 – ECOLOGY (1.2): FINAL EXPLANATION

Student Assessment Document

Student Name	_____
Class	_____
Date	_____

INSTRUCTIONS FOR STUDENTS

Driving Question: Why are fish disappearing from a water body that once supported many fish, and what can the community do to restore the ecosystem? You have watched the Kenyan river video (2:20 to 4:50), investigated habitats, adaptations, populations, sampling methods, limiting factors, nutrient cycles and pollution. Now write your own Final Explanation. Answer each section in the space under the prompt, using evidence from your lessons, your presentations, your reading summary, your cycle drawings and your pollution summary. Use correct ecological terms and Kenyan examples. Write in full sentences. Where asked, you may add a labelled sketch or a simple calculation.

Section 1: The River as an Ecosystem

At the start, people thought the fish were old or had no food. (a) Using what you learned in Lesson 1, explain why these ideas are not enough to explain what is happening in the river. (b) Use at least four ecological terms (for example habitat, population, community, niche, biotic factor, abiotic factor) to describe the river. (c) Explain what the phrase 'the river is still full of life' tells you about how organisms interact with their environment.

Write your answer here:

Section 2: Adaptations and Habitats

(a) Choose two organisms from different habitat groups that you studied (for example Euphorbia, maize, water hyacinth, mangroves, camel, tilapia). For each, name its habitat group and describe one adaptation and how it helps the organism survive. **(b)** Water hyacinth and algae both grow thickly in the river. Explain which of them is a hydrophyte and which is not. **(c)** Explain why conditions in the river can favour weeds and algae but harm fish. Link your answer to the adaptations of fish.

Write your answer here:

Section 3: Measuring Populations

(a) Define population and population density. (b) In the school compound your group counted every individual in a 10 m x 10 m area. Explain when direct counting is suitable and why it would be difficult in the river. (c) Describe how quadrat sampling (1 m x 1 m quadrats placed randomly) and a belt transect helped you find out whether plants are spread evenly. State what you found and what caused the differences. (d) In a capture-recapture simulation, 20 counters were marked and returned to the container. A second sample of 20 contained 5 marked counters. Calculate the estimated population and compare it with the true value of 100. State two assumptions of the method and say how the method could help the community know if fish numbers are falling.

Write your answer here:

Section 4: Why Are the Fish Disappearing?

Use your reading summary on limiting factors, your nitrogen and carbon cycle drawings, and your pollution summary. (a) Name two biotic and two abiotic factors that can limit a population, and explain what carrying capacity means. (b) Describe how nitrogen and nutrients from a place such as a maize shamba, a dairy farm near Kericho or sewage can reach the river. (c) Write a step-by-step explanation, in order, of how this leads to the booming of algae and weeds, the bad smell and the loss of fish. (d) Explain how garbage and effluent, as in the Nairobi and Athi Rivers, add to the problem.

Write your answer here:

Section 5: Restoring Our River

The driving question asks what the community can do to restore the ecosystem. (a) Propose three actions the community could take, and for each explain which part of the problem it addresses. (b) Describe one way you could check, using a method from this unit, whether your actions are working. (c) Compare your first model of the river in Lesson 1 with what you now know. State two things you would add or change.

Write your answer here:

RUBRIC			
Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Use of evidence from the lessons	Uses specific evidence from the river video, the school field investigations, simulations, readings and pollution research to support every claim.	Uses some evidence from the lessons, but some claims are general or unsupported.	Gives few or no pieces of evidence, or evidence is not linked to the river.
Scientific accuracy and ecological terms	Uses terms such as ecosystem, habitat, population, density, carrying capacity, eutrophication and dissolved oxygen correctly, with accurate adaptations and calculations.	Uses most terms correctly with minor errors in adaptations or calculations.	Uses few terms or makes major errors, such as confusing algae with hydrophytes or population with community.
Causal explanation of the phenomenon	Gives a complete ordered chain from nutrient runoff to algal bloom, decomposition, low oxygen, smell and fish loss, and mentions other limiting factors.	Gives a mostly correct chain but misses a step or does not link the smell or oxygen.	Lists causes without explaining how they connect, or relies on the original ideas (old fish, no food).

Sampling methods and data	Explains direct counting, quadrats, belt transects and capture-recapture correctly, calculates the estimate accurately and states assumptions.	Describes methods and calculates correctly, but explanations of when to use them or the assumptions are incomplete.	Confuses methods or cannot calculate the estimate.
Community solutions and model revision	Proposes realistic actions that each target a cause, suggests a way to monitor results, and clearly shows how the model has improved.	Proposes reasonable actions and some model changes, but links to causes or monitoring are weak.	Gives vague or unrelated actions and little model revision.

ATTRIBUTION AND LICENSING

Curriculum Alignment: Structured in pedagogical alignment with the competence-based curriculum format and learning outcomes established by the Kenya Institute of Curriculum Development (KICD). This material is independently developed and is not officially endorsed or published by KICD.

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